$$X_2 = \mu_2 + l_{21}F_1 + l_{22}F_2 + \cdots + l_{2m}F_m + e_2$$

$\vdots$

$$X_p = \mu_p + l_{p1}F_1 + l_{p2}F_2 + \cdots + l_{pm}F_m + e_p$$

$$\underset{p\times 1}{\tilde{X}} = \underset{p\times 1}{\mu} + \underset{p\times m}{L}\,\underset{m\times 1}{\tilde{F}} + \underset{p\times 1}{e}$$

unexplainable factors specific to Feature

↳ Loading matrix

<u>Assumptions</u> $E(F_i) = 0$, $var(F_i) = 1$, $cov(F_i, F_j) = 0$ $\Big\}$ $E(F) = 0$, $var\text{-}cov(F) = I_m$ (also called orthogonal factor model)

$E(\tilde{e}) = 0$, $cov(\tilde{e}) = diag(\psi_1, \psi_2, \ldots, \psi_p)$

$cov(e_i, e_i) = \psi_i$; $cov(\psi_i, \psi_j) = 0$; $i \neq j$

J clan mined

$$R = \Sigma = \begin{bmatrix} 1 & .63 & .45 \\ .63 & 1 & .35 \\ .45 & .35 & 1 \end{bmatrix}$$

$m = 1$, $L = [.9 \ \ .7 \ \ .5]^T$

$$LL^T = \begin{bmatrix} .81 & .63 & .45 \\ .63 & .49 & .35 \\ .45 & .35 & .25 \end{bmatrix}$$

$$\Psi = \begin{bmatrix} .19 & & 0 \\ & .51 & \\ 0 & & .75 \end{bmatrix}$$

If $m = p$; $\forall \Sigma$ factor ~~decom~~ decomposition exists

$\Sigma_{p \times p} \to$ symm pd

$\Sigma = P \Lambda P^T$

$= \sum_{i=1}^{p} \lambda_i e_i e_i^T$

$$= [\sqrt{\lambda_1} e_1 \ \sqrt{\lambda_2} e_2 \cdots \sqrt{\lambda_p} e_p]_{p \times p} \begin{bmatrix} \sqrt{\lambda_1} e_1^T \\ \sqrt{\lambda_2} e_2^T \\ \vdots \\ \sqrt{\lambda_p} e_p^T \end{bmatrix}_{p \times p}$$

$L = [\sqrt{\lambda_1} e_1 \cdots \sqrt{\lambda_p} e_p]_{p \times m}$ $\quad m = p$

~~to~~ We get cor. of factors as I

For $m<p$, ~~some~~ factor model can be fit only for some Σ

$L: p \times p$

$\underset{\sim}{X} = \mu + L \underset{\sim}{f} + \underset{\sim}{e}$

Estimate by sample — p

Estimate by sample data — p^2

$\underset{\sim}{X} \to \mu \; \Sigma$ — symm p.d.

$p \qquad \frac{p(p+1)}{2}$

$\frac{p(p+3)}{2}$

$p(p+1)$

$\frac{p(p+1)}{2} \qquad \frac{p(p+3)}{2}$

$p^2 > \frac{p(p+1)}{2}$

An issue, defeating the idea of factor model

We always want $m<p$ & m is least the integer for which ~~faster~~ factor model exists

• Even if $\Sigma = LL^t + E$ for a given $m<p$ exists is L unique?

Ans: L is unique upto orthogonal transformation. (orthonormal)

$\underset{p \times m}{L^*} = L \underset{m \times m}{T}$. T is an orthogonal matrix

$TT^t = TT^t = I_m$ ~~diagonal matrix~~

$\underset{\sim}{X} = \mu + Lf + e = \mu + LTT^t f + e$

$= \cancel{\mu + L^* F + e} = \mu + L^* (\underset{\sim}{F^*}) + e$

$F^* = T^t f$ — new m-unknown factors.

$F^* = T^t F$

$(F_1 \dots F_m)$ $\quad$ $(F_1^* \dots F_m^*)$

X $\quad$ X

$E(F^*) = T^t E(F) = 0$

In new model

$$Cov(\underset{\sim}{X}) = L^* L^{*t} + \Psi$$
$$= LT(LT)^t + \Psi$$
$$= LTT^t L^t + \Psi$$
$$= LL^t + \Psi$$

$$Cov(F^*) = Cov(T^t F)$$
$$= T^t \, Cov(F) \, T$$
$$= T^t T \, Cov(F)$$
$$= I_m \, Cov(F)$$
$$= Cov(F) = I_m$$

Estimating factor model from sample

1. Principal Component Solution / method
2. Maximum Likelihood Solution. / method

1. PCS

$$\Sigma = P \Lambda P^T \quad \lambda_1 \geq \lambda_2 \dots \geq \lambda_p$$
$$= [\sqrt{\lambda_1} e_1 \cdots \sqrt{\lambda_p} e_p] \begin{bmatrix} \sqrt{\lambda_1} e_1 \\ \vdots \\ \sqrt{\lambda_p} e_p \end{bmatrix}$$

$$= \sum_{i=1}^{p} \lambda_i e_i e_i^T = \sum_{i=1}^{m} \lambda_i e_i e_i^T + \sum_{i=m+1}^{p} \lambda_i e_i e_i^T$$

$$= L_m L_m^T + L_{m+1} L_{m+1}^T$$

$$L = L_m = [\sqrt{\lambda_1}\, e_1 \cdots \sqrt{\lambda_p}\, e_m]_{p\times m}$$

$$L_{m+1} = [\sqrt{\lambda_{m+1}}\, e_{m+1} \cdots \sqrt{\lambda_p}\, e_p]$$

$$\Sigma = LL^T + \Psi$$

Taking $L = L_m$

$$\Psi = \underbrace{\text{diag}}_{\text{diagonal elements}}\left(L_{m+1} L_{m+1}^T\right)$$

Although $L_{m+1}L_{m+1}^T$ has off-diagonal non-zero elements but assuming them to be too small to get neglected.
We are committing error in ~~calculating~~ fitting $\Sigma = LL^T + \Psi$

$$\underbrace{L_m L_{m+1}^T}_{\text{symm}} = \begin{bmatrix} \psi_1 & & 0 \\ & \ddots & \\ 0 & & \psi_p \end{bmatrix} + \begin{bmatrix} 0 & & \text{offdiagonal} \\ & \ddots & \\ & & 0 \end{bmatrix}$$

Residue matrix
Error ~~comm~~ committed

Frobenius of this matrix $< \epsilon$ (tolerance) then we have the model

$A = [a_{ij}]$

$\|A\|_F = \sqrt{\sum\sum a_{ij}^2}$ Frobenious Norm

Take $\boxed{m}$ s.t. $\dfrac{\lambda_1 + \cdots + \lambda_m}{\lambda_1 + \cdots + \lambda_p}$ is good enough to capture population variance

Or use scree plot